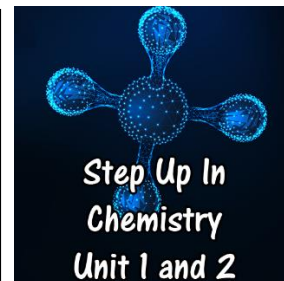


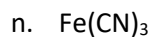
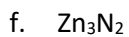
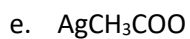
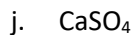
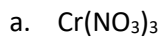
2.2 Ionic Bonding

Problems Worksheet



1. Sodium metal reacts with chlorine gas.
 - a. Explain what occurs when sodium metal reacts with chlorine gas.
 - b. Name and write the formula for the compound produced.
 - c. Draw a Lewis structure for the compound.
 - d. Describe the structure of the compound formed. Include a diagram.
2. Draw diagrams showing the formation of the following ionic compounds:
 - a. Lithium fluoride
 - b. Magnesium sulfide

3. Write the name for the following ionic compounds:



4. Write the formula for the following ionic compounds:

a. Sodium hydroxide

h. Nickel nitrate

b. Tin(II) fluoride

i. Aluminium sulfite

c. Copper(II) hydrogencarbonate

j. Silver iodide

d. Zinc phosphate

k. Iron(II) chromate

e. Strontium permanganate

l. Magnesium hydrogenphosphate

f. Iron(III) cyanide

m. Cobalt(II) nitrite

g. Lead(II) oxalate

n. Chromium nitride

5. The table to the right shows the melting point of a number of different ionic compounds. Explain the melting points of ionic compounds.

Compound	Melting Point (°C)
Potassium bromide	734
Sodium chloride	801
Silver nitrate	212

6. Ionic compounds are brittle and not malleable like metals. Explain why ionic compounds are brittle. You should include a diagram in your answer.

7. Nic and Owen decide to do an experiment on the electrical conductivity of ionic compounds in different states using a circuit, battery, electrodes and an ammeter.
- Draw a diagram of how they might set up their experiment.
 - They decided to use silver nitrate in their experiment instead of sodium chloride. Why?
 - They test the conductivity of their silver nitrate salt as a solid, liquid and as a solution in water. Predict the results they would have seen and explain the results.